HW4: Multimodal AI – Visual Instruction Tuning (VQA)

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1. Title Page

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GitHub: https://github.com/fader2077/Recurrent-Neural-Network-and-Transformer-HW/tree/main/hw4

2. Overview

This report preserves the original HW4 required experiment and extends it with controlled re-evaluation, conservative metrics, and stricter qualitative checks.

3. Dataset Choice

ChartQA is selected because chart VQA requires numeric reading, legend grounding, color recognition, and trend reasoning.

4. Model and Method

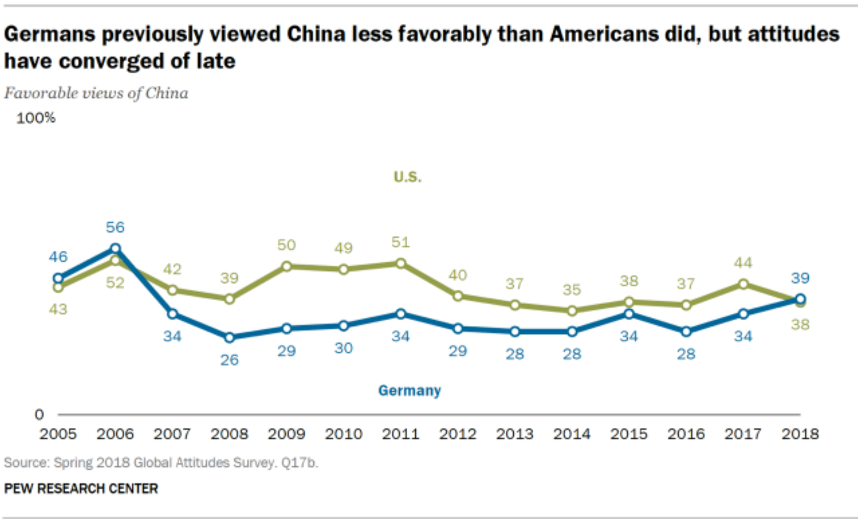
Base model: LLaVA-v1.5-7B. Quantization: 4-bit NF4. Fine-tuning method: QLoRA / PEFT LoRA.

5. Training Setup

Original required setup is preserved. The official v2 run uses a custom multimodal Trainer/PyTorch-style loop with a custom collator, assistant-only loss masking (labels for answer tokens only), padding labels=-100, train[:1000], 3 epochs, q/v LoRA targets, learning rate 1e-4, batch size 1, gradient accumulation 16, 4-bit NF4, frozen vision tower and frozen projector, and no truncation.

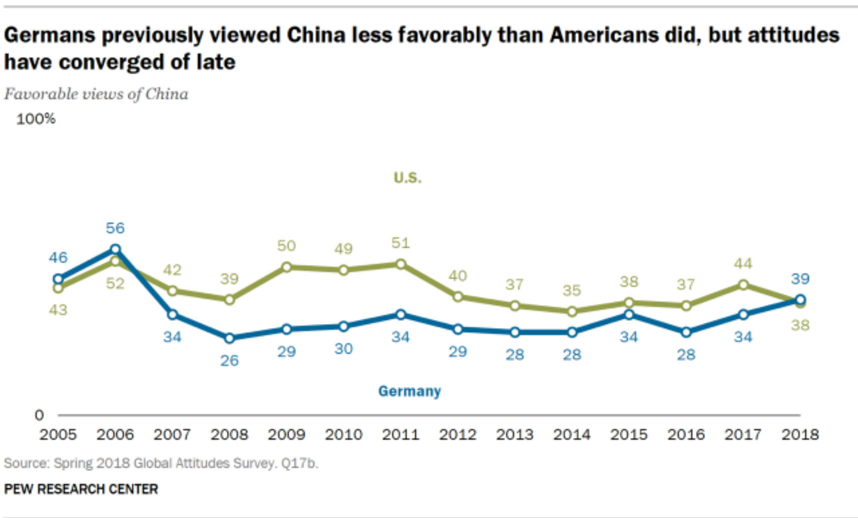
6. Original 5-Case Evaluation

Case 0



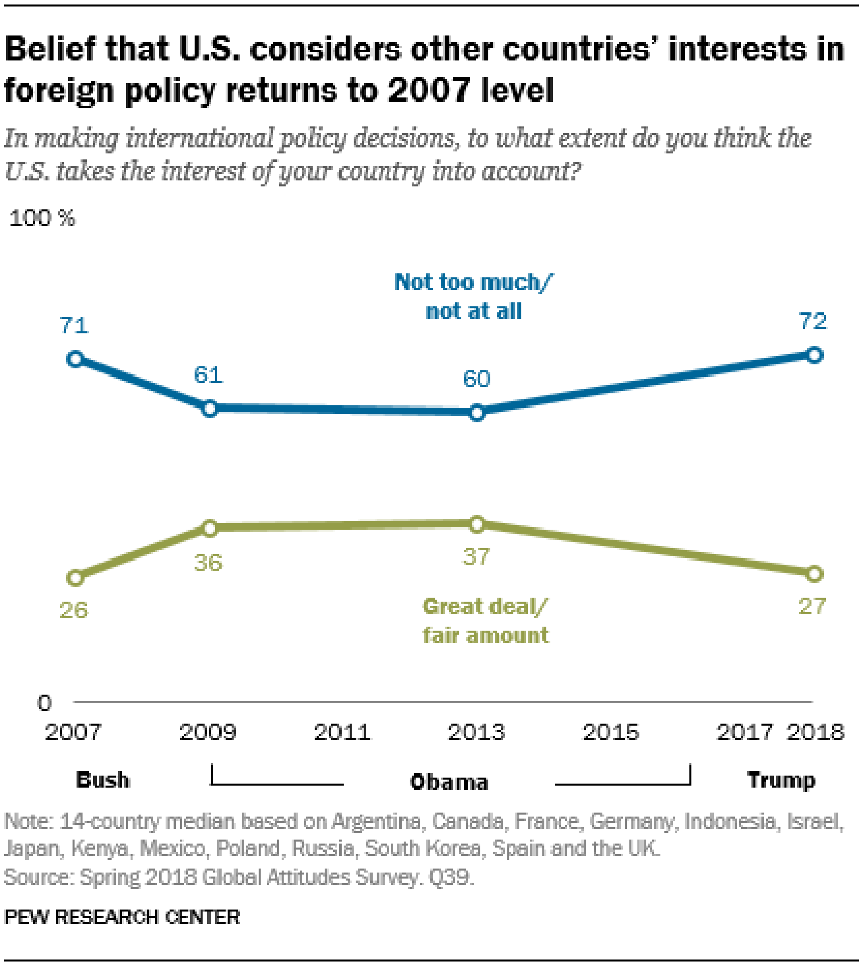
Question: What's the color of graph with 56 as the highest value?
Ground Truth: Blue
Base Model Answer: The color of the graph with 56 as the highest value is blue.
Fine-tuned Answer: Blue
Analysis: Both models identify the correct color. The base model gives a verbose sentence, while the fine-tuned adapter returns the concise label.

Case 1



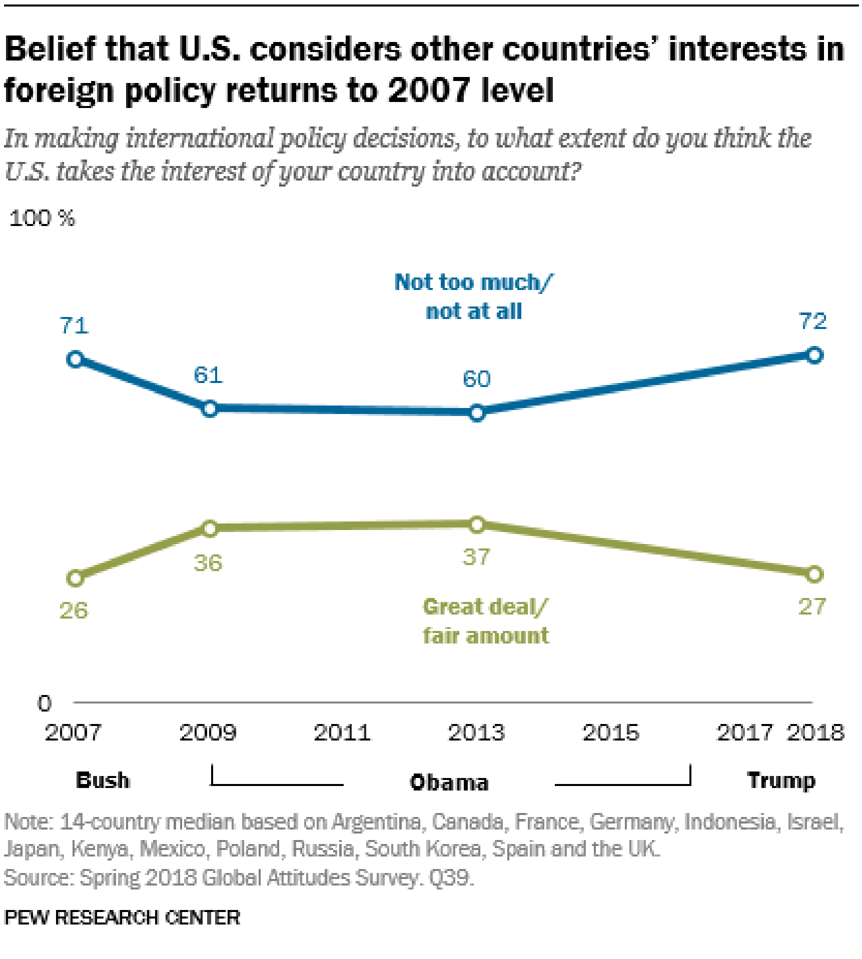
Question: In which year the difference between blue and green graph 1?
Ground Truth: 2018
Base Model Answer: The difference between the blue and green graphs in 2011 is 1.
Fine-tuned Answer: 2005
Analysis: The base model predicts 2011 and the adapter predicts 2005; both miss the ground-truth year 2018.

Case 2



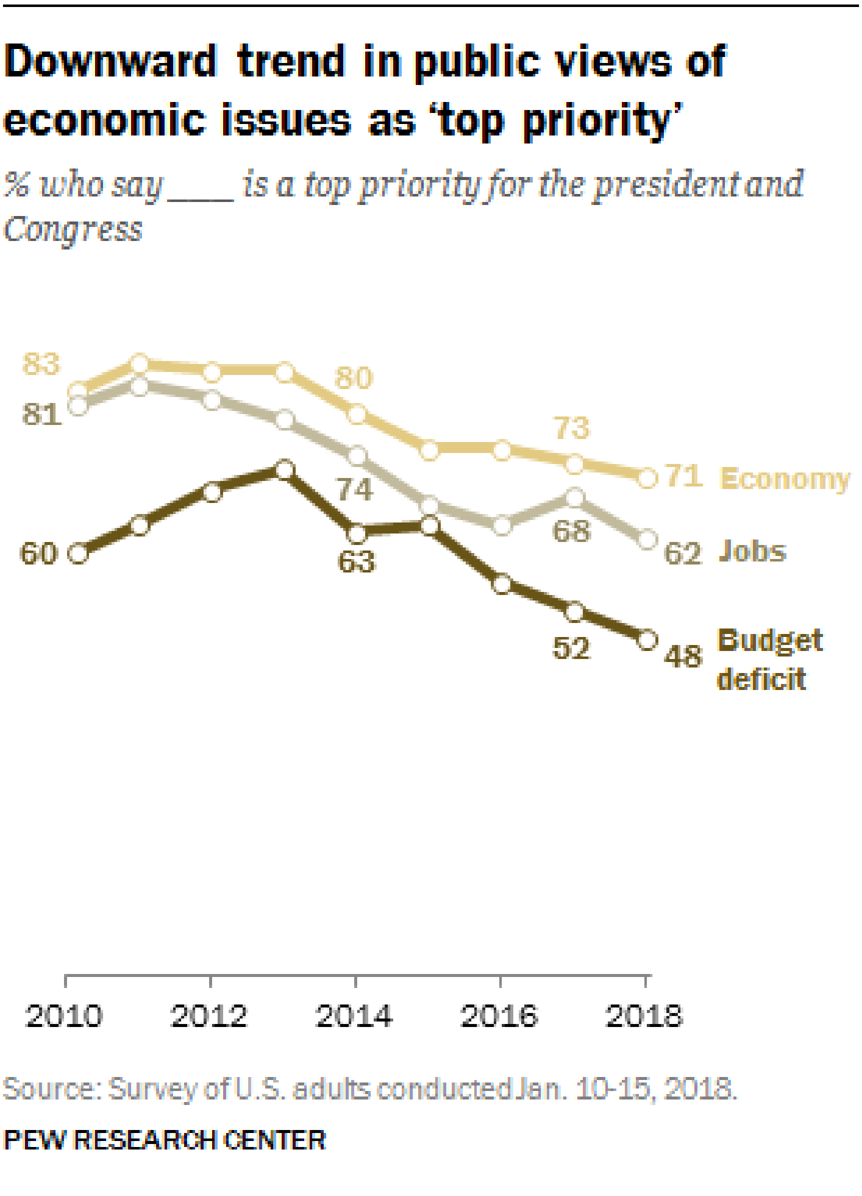
Question: What does the blue line represent?
Ground Truth: Not too much/not at all
Base Model Answer: The blue line represents the percentage of foreign policy decisions made by the United States.
Fine-tuned Answer: The blue line represents the value of the foreign policy.
Analysis: Both models miss strict legend grounding and do not return the exact chart label.

Case 3



Question: What is the max value of blue line?
Ground Truth: 0.72
Base Model Answer: The max value of the blue line is 70%.
Fine-tuned Answer: 1.0
Analysis: The base model predicts 70% and the adapter predicts 1.0; both miss the ground truth 0.72.

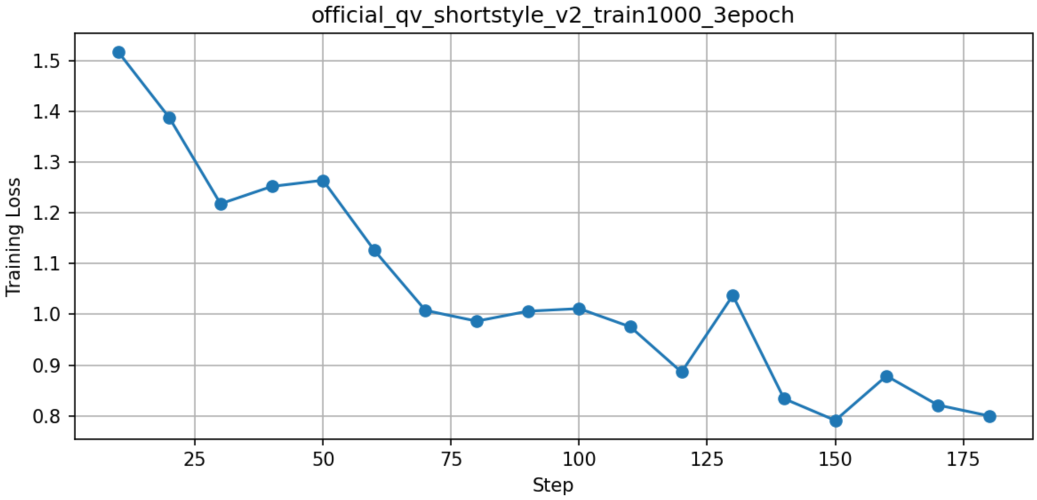
Case 4



Question: What's the percentage of respondents who say Job is a top priority for the president and Congress in 2016?
Ground Truth: 68
Base Model Answer: In 2016, 67% of respondents indicated that jobs were a top priority for the president and Congress.
Fine-tuned Answer: 75.75
Analysis: The base model predicts 67 and the adapter predicts 75.75; both miss 68.

7. Original Loss Curve

Legacy diagnostic: the earlier curve (outputs/loss_curve.png) started near 10 and is treated as suspicious due to likely label-masking issues. **Official clean training evidence:**



The official v2 run starts around 1.52 and decreases to around 0.80, with final train loss about 1.033. This is substantially cleaner than the legacy high-loss curve and supports the corrected masking pipeline.

8. Additional Experiments Overview

Primary rule: fix evaluation logic and reporting consistency first, then interpret improvements conservatively.

9. Larger Evaluation on ChartQA val[:50]

Model	Exact	Norm	NumNear	CombinedRelaxed
Base	0.000	0.180	0.100	0.240
q/v 1ep	0.060	0.060	0.140	0.180
q/v 2ep	0.080	0.120	0.120	0.220
q/k/v/o 1ep	0.060	0.060	0.100	0.140

Combined Relaxed Accuracy is a relaxed metric, not strict accuracy. Base LLaVA has very low exact match in the normal prompt setting because it often produces verbose sentence-style answers instead of short ChartQA labels.

10. Strict vs Relaxed Metrics

Keep strict and relaxed interpretations separate: case studies use strict ground-truth comparison, while relaxed metrics capture near-miss behavior.

11. Controlled Re-evaluation on ChartQA val[:200]

Generation configuration: do_sample=False, num_beams=1, max_new_tokens=16, repetition_penalty=1.1.

Model	Prompt	Exact	Norm	FirstNum	AnyNum	Cons	Relaxed
Base	normal	0.005	0.130	0.090	0.090	0.200	0.200
Base	short	0.115	0.125	0.210	0.210	0.290	0.290
Base	dataset	0.130	0.135	0.210	0.210	0.305	0.305
q/v 1ep	normal	0.125	0.135	0.200	0.200	0.300	0.300
q/v 1ep	short	0.110	0.120	0.220	0.220	0.310	0.310
q/v 1ep	dataset	0.110	0.110	0.225	0.225	0.305	0.305
q/v 2ep	normal	0.120	0.130	0.195	0.195	0.285	0.285
q/v 2ep	short	0.100	0.110	0.190	0.190	0.275	0.275
q/v 2ep	dataset	0.095	0.110	0.175	0.175	0.260	0.260
q/k/v/o 1ep	normal	0.110	0.115	0.185	0.185	0.275	0.275
q/k/v/o 1ep	short	0.100	0.110	0.195	0.195	0.285	0.285
q/k/v/o 1ep	dataset	0.100	0.105	0.175	0.175	0.255	0.255
q/v clean 3ep	normal	0.080	0.285	0.265	0.275	0.460	0.460
q/v clean 3ep	short	0.055	0.255	0.245	0.255	0.430	0.435
q/v clean 3ep	dataset	0.025	0.250	0.245	0.250	0.425	0.425
q/k/v/o clean 2ep	normal	0.105	0.230	0.250	0.255	0.400	0.405
q/k/v/o clean 2ep	short	0.105	0.230	0.255	0.260	0.415	0.420
q/k/v/o clean 2ep	dataset	0.085	0.225	0.265	0.270	0.420	0.425

12. Conservative vs Relaxed Metrics

Combined Conservative Accuracy (Cons) is the main metric after observing unstable long numeric strings. Combined Relaxed Accuracy (Relaxed) is retained as an upper-bound relaxed metric.

13. Answer Stability Diagnostics

Model	Prompt	MultiNum	Overlong	Repeat	AvgLen	Verbose
Base	normal	0.000	0.140	0.000	11.105	0.955
Base	short	0.000	0.000	0.000	1.280	0.035
Base	dataset	0.000	0.000	0.000	1.140	0.015
q/v 1ep	normal	0.000	0.005	0.000	1.800	0.075
q/v 1ep	short	0.000	0.000	0.000	1.085	0.005
q/v 1ep	dataset	0.000	0.000	0.000	1.065	0.000
q/v 2ep	normal	0.000	0.000	0.110	1.415	0.060
q/v 2ep	short	0.000	0.000	0.080	1.065	0.000
q/v 2ep	dataset	0.000	0.000	0.120	1.035	0.000
q/k/v/o 1ep	normal	0.000	0.000	0.000	1.570	0.075
q/k/v/o 1ep	short	0.000	0.000	0.000	1.085	0.010
q/k/v/o 1ep	dataset	0.000	0.000	0.000	1.010	0.000
q/v clean 3ep	normal	0.035	0.015	0.430	2.870	0.260
q/v clean 3ep	short	0.035	0.005	0.415	2.835	0.270
q/v clean 3ep	dataset	0.050	0.000	0.450	2.805	0.290
q/k/v/o clean 2ep	normal	0.020	0.005	0.425	2.105	0.160
q/k/v/o clean 2ep	short	0.025	0.010	0.395	1.940	0.115
q/k/v/o clean 2ep	dataset	0.010	0.005	0.410	1.925	0.140

Base LLaVA with normal prompt is verbose. Short prompts reduce verbosity. Clean adapters improve relaxed metrics but still show non-trivial repeated-fragment rates.

14. Label Masking Diagnostics

Diagnostics file: `outputs/additional_experiments/label_masking_diagnostics.txt`. It prints question, ground truth answer, decoded full input, decoded label tokens where labels != -100, and masked ratio.

15. Numeric Near-Miss Analysis

Model	Total Numeric Questions	Strict Exact Correct	Numeric Near-Match Correct	Strict Wrong but Numeric Near-Match Correct	Numeric Near-Match Rate
Base LLaVA	32	0	4	4	0.125
Original adapter q/v 1 epoch	32	1	5	4	0.156
New adapter q/v 2 epochs	32	2	4	3	0.125
New adapter q/k/v/o 1 epoch	32	1	3	2	0.094

Case studies are discussed under strict ground-truth comparison, while numeric tolerance is used only as a relaxed aggregate metric.

16. Clean Retraining Results

Model	Exact	Norm	FirstNum	AnyNum	Cons	Relaxed
q/v clean 3ep	0.080	0.285	0.265	0.275	0.460	0.460
q/k/v/o clean 2ep	0.105	0.230	0.250	0.255	0.400	0.405

Clean retraining improves relaxed aggregate metrics, especially normalized matching and numeric near-match. Exact match remains much lower than relaxed metrics.

17. Controlled Final Clean Adapter 5-Case Check

Main table shows shortened outputs (max 60 chars); full outputs are saved to CSV/JSON. For official v2, the three repeated rows for each case correspond to normal, short-answer, and dataset-style prompts.

Case	Model	Prompt	GroundTruth	Answer
0	Base	normal	Blue	The color of the graph with 56 as the highest value is blue.
0	official v2	normal	Blue	Green
0	official v2	short	Blue	Green
0	official v2	dataset	Blue	Green
0	q/v 1ep	normal	Blue	Blue
1	Base	normal	2018	In the year 2008, there is a noticeable difference in
1	official v2	normal	2018	2013
1	official v2	short	2018	2008
1	official v2	dataset	2018	2008
1	q/v 1ep	normal	2018	2005
2	Base	normal	Not too much/not at all	The blue line represents the percentage of foreign policy de...
2	official v2	normal	Not too much/not at all	The blue line represents the “not too much/not at all” optio...
2	official v2	short	Not too much/not at all	Not too much/Not at all
2	official v2	dataset	Not too much/not at all	Not too much/Not at all.
2	q/v 1ep	normal	Not too much/not at all	The blue line represents the value of the foreign policy.
3	Base	normal	0.72	The maximum value of the blue line in the image is at around...
3	official v2	normal	0.72	The maximum value of the blue line, which represents “not to...
3	official v2	short	0.72	70.1428571429 (7
3	official v2	dataset	0.72	70.526315794844
3	q/v 1ep	normal	0.72	1.0
4	Base	normal	68	In 2016, 74% of respondents indicated that
4	official v2	normal	68	71.43%
4	official v2	short	68	71.43%
4	official v2	dataset	68	71.43%
4	q/v 1ep	normal	68	57.3%

Although the official v2 retraining improves aggregate conservative metrics versus the original q/v 1ep model on val[:200], Case 0 still regresses from Blue to Green. Therefore, strict chart reasoning is still not solved and this result is reported as a limitation.

18. Format-Control Evaluation

Short-answer and dataset-style prompts reduce verbosity strongly, but correctness gains are smaller than format gains. Official v2 summaries are saved to `outputs/official_qv_shortstyle_v2_train1000_3epoch/summary.csv`, `outputs/official_qv_shortstyle_v2_train1000_3epoch/val200_eval.csv`, and `outputs/official_qv_shortstyle_v2_train1000_3epoch/fixed_5case_eval.csv`.

19. Statistical Caution

Because the evaluation subset contains 200 examples, a 0.01 change corresponds to about two examples. Therefore, small differences should not be over-interpreted.

20. Final Discussion

The QLoRA pipeline satisfies the HW4 requirements and trains successfully. The original five-case evaluation shows persistent failures in exact year extraction, legend grounding, and numeric reading. The old high-loss curve is treated as suspicious, and the official clean shortstyle retraining provides cleaner training evidence with plausible loss dynamics. On val[:200], the official v2 adapter improves aggregate metrics versus the original q/v 1ep baseline under controlled generation, and Combined Conservative Accuracy is treated as the primary metric. However, exact-match and strict chart reasoning remain limited, and Case 0 still regresses to Green. Therefore, conclusions remain conservative.

21. Conclusion

Small differences should not be over-interpreted. On val[:200], a 0.01 difference corresponds to about two examples. The final claim is limited: fine-tuning improves format control and aggregate conservative metrics, but strict chart reasoning remains difficult.